

Highlights

Dual-Field Electricity Price Forecasting with Predispatch

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- A trend–event dual-field model is adapted from reconstruction to price forecasting.
- Static-fit fields worsen MAE but improve CRPS relative to the same linear base.
- Quarterly refits and predispatch yield pooled gains, with regional differences.
- Negative-price scores improve over all five main-table comparators.
- A validation-selected lower bound trades tail accuracy for lower overall MAE.

Dual-Field Electricity Price Forecasting with Predispatch

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Abstract

Wholesale electricity prices combine a changing level with scarcity spikes and negative-price troughs. We adapt the trend–event structure of DualTimesField to rolling 24-hour forecasting using detected historical events, a linear base, and point-in-time market-operator forecasts. Experiments cover three regions of the Australian National Electricity Market, with 17,521 hourly origins per region in 2023–2024. On a single fit to 2015–2021, the fields worsen mean absolute error (MAE) relative to the same linear base, although they improve its approximate continuous ranked probability score (CRPS). With quarterly refits and AEMO predispatch inputs, the fields improve both metrics when pooled across regions; point-forecast gains are not uniform across regions. A shared per-hour calibrator and a validation-selected lower bound further reduce MAE. The final model achieves a three-region mean MAE of 36.6 AUD/MWh, below PatchTST, DLinear and a linear baseline (pooled Diebold–Mariano $p < 0.01$), but above XGBoost at 35.5. Its negative-price CRPS is lower than all five main-table comparators ($p < 0.001$). The lower bound trades a small deterioration in negative-price scores for better overall MAE. These results identify the conditions and limits of the decomposition’s forecasting value.

Keywords: Electricity price forecasting, Probabilistic forecasting, Time-series decomposition, Negative prices, National Electricity Market, Predispatch

1. Introduction

Short-term electricity price forecasts drive bidding, storage dispatch, demand response and hedging decisions [1, 2]. In markets with a fast-growing

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4 share of wind and solar generation, these forecasts have become harder. Prices
5 still follow a daily and seasonal level, but they are increasingly punctuated
6 by events: scarcity spikes when supply margins tighten and negative-price
7 troughs when renewable output exceeds demand. In the Australian National
8 Electricity Market (NEM), the share of hours with negative prices in Queens-
9 land rose from 1.6% in 2015–2021 to 13.2% in 2023–2024 (Section 3). A
10 forecaster that treats these events as noise around a level misses exactly the
11 hours in which storage and flexible demand earn or lose most.

12 Separating a series into a continuous trend and discrete events is therefore
13 an attractive structural prior. Classical decompositions and their neural
14 successors [3, 4, 5] split trend from seasonality; DualTimesField [6] goes
15 further and represents a series as a continuous trend field (CTF), an implicit
16 neural representation in continuous time [7, 8], plus a discrete event field
17 (DGF) of localized atoms. It was designed for reconstruction and interpolation.
18 Whether such a representation helps forecasting is not obvious: describing
19 the past more faithfully does not by itself add information about the future,
20 and a richer description of a past regime can overfit it.

21 This paper studies that question for rolling 24-hour price forecasting in
22 three NEM regions, under a same-data protocol in which every compared
23 model receives the same underlying inputs, the same training periods and the
24 same evaluation. Our findings are sequential, and the paper follows them:

- 25 1. *Transplanted, the dual field does not separate events.* The CTF absorbs
26 spikes and the DGF collapses. Detecting events explicitly, as the
27 largest departures from the window median, restores the separation
28 (Appendix A).
- 29 2. *Separation alone does not improve point accuracy.* On a single fit to
30 2015–2021, a linear model on the same inputs has lower MAE than
31 the dual-field model. Adding the fields to the same linear base raises
32 MAE but lowers CRPS (Section 6.2). Lower training loss and higher
33 validation MAE are consistent with overfitting to the training-period
34 price regime.
- 35 3. *Recalibration and market-operator information yield pooled gains.* Moti-
36 vated by the frequent recalibration used in electricity price forecasting
37 [9], we refit every model before each test quarter. AEMO’s predispach
38 combines current offers, constraints and demand forecasts, and lowers
39 every model’s three-region mean MAE. With these inputs and refits,
40 the dual fields improve pooled MAE and CRPS over the same linear

base; raw-price MAE nevertheless worsens in QLD1.

4. *Diagnosis-driven corrections.* A per-region analysis of the errors that remain against the strongest competitor locates them in negative-price hours, where linear heads over-extrapolate in the transformed target space. A shared per-hour calibrator and a price floor chosen on validation data address this and are each significant.

The final model has lower MAE than PatchTST [10], DLinear [5] and a linear model, and lower negative-price CRPS and lower-tail pinball loss than the five main-table comparators. XGBoost [11] with the same inputs and refits remains more accurate on overall MAE and CRPS. The final model is not the best of all development variants on negative-price scores: omitting its lower bound improves those scores but worsens overall MAE. Beyond the model, the study shows why a structured forecaster should be evaluated against strong same-input baselines and under its intended recalibration regime.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data and the forecasting task, Section 4 the model, and Section 5 the experimental protocol. Section 6 reports the results, Section 7 discusses them, and Section 8 concludes.

2. Related work

Electricity price forecasting. Reviews by Weron [1] and Nowotarski and Weron [12] survey point and probabilistic electricity price forecasting. Lago et al. [9] established an open benchmark and best practices, among them daily recalibration on a rolling window, variance-stabilizing transformations [13], and comparison against well-specified linear (LEAR) and neural baselines with Diebold–Mariano tests [14]. High-dimensional linear models [15] and distributional neural networks [16] are strong references. We follow these practices: an asinh transformation, rolling recalibration, same-input baselines, and significance tests.

Deep time-series forecasting and decomposition. Transformer forecasters such as Informer [17], Autoformer [3], PatchTST [10] and iTransformer [18], recurrent models [19, 20] and DeepAR [21] have been widely applied. Zeng et al. [5] showed that a decomposed linear model (DLinear) is competitive with many of them. N-BEATS [4] and Autoformer use trend and seasonal decompositions inside the network; hybrid models combine a linear component

with a nonlinear residual [22]. DualTimesField [6] represents trend and events as continuous-time fields for reconstruction. Our model adds dual fields as a residual to a linear base, and asks when this structure pays off in forecasting.

Exogenous information and strong tabular baselines. Fundamental forecasts of load, renewable output and capacity are established inputs in electricity price forecasting [1, 9]. Recent work adds text: Chen et al. [23] propose a reasoning-enhanced probabilistic electricity price forecaster using a parameter-efficient large language model. Our study uses numerical market-operator forecasts; RE-Price is related work, not a reproduced same-data competitor. The strong performance of gradient-boosted trees on tabular benchmarks [24] motivates including XGBoost alongside neural baselines.

3. Data and forecasting task

Market and prices. We use hourly regional reference prices (RRP) and operational demand for New South Wales (NSW1), Queensland (QLD1) and Tasmania (TAS1) from AEMO’s aggregated price and demand data [25], 1 January 2015 to 31 December 2024 in market time (AEST), with half-hourly records averaged to hours. Table 1 summarizes the series. The test period, 2023–2024, differs markedly from 2015–2021: negative prices are four to eight times more frequent, spikes above 300 AUD/MWh are more frequent in NSW1 and QLD1, and 2022, the energy-crisis year that separates them, has a much higher level.

Forecasting task. Let p_t denote the price for the delivery hour starting at t . At origin t , the target is $(p_t, \dots, p_{t+23})$, using observations from hours $t - 72, \dots, t - 1$ and exogenous information published by t . The observation for hour $t - 1$ becomes available at t ; no target-hour observation enters the history. We index the 24 forecast outputs by $h = 0, \dots, 23$. Origins advance hourly, giving 17,521 complete test windows per region. This is a rolling 24-hour task, distinct from a fixed-origin day-ahead auction forecast.

Known inputs. Four groups of forward-looking information enter every model, each as published at the origin:

- *Calendar:* hour of day, day of week and day of year (sine and cosine), month, weekend flag and a standardized year.

Table 1: Hourly regional reference prices (AUD/MWh) by period. Neg.: share of hours below zero; > 300: share of hours above 300 AUD/MWh.

Region	Period	Mean	Median	Std. dev.	Neg.	> 300
NSW1	2015–2021	70.5	58.0	164.9	0.37%	0.25%
	2022	182.7	135.4	215.3	2.61%	14.79%
	2023–2024	113.5	82.7	408.8	6.20%	1.75%
QLD1	2015–2021	70.9	55.2	202.5	1.63%	0.68%
	2022	205.1	139.9	456.2	4.05%	15.73%
	2023–2024	101.1	81.3	289.2	13.21%	1.85%
TAS1	2015–2021	69.4	57.6	68.7	2.53%	0.43%
	2022	147.0	104.9	285.3	1.82%	8.57%
	2023–2024	75.8	60.3	168.2	7.36%	0.50%

- *PD PASA*: AEMO’s pre-dispatch projected assessment of system adequacy [26], from which we take maximum spare capacity and net load (50%-probability demand minus semi-scheduled wind and solar availability) for each forecast hour, from the latest run published by the origin, with the short-term PASA run filling hours beyond its horizon.
- *Gas*: the logarithm of the seven-day trailing mean of the Victorian declared wholesale gas market price [27], using only gas days ended before the origin.
- *Predispatch*: AEMO’s predispatch re-solves the market dispatch every 30 minutes with current offers, constraints and demand forecasts. Its horizon extends to the end of the trading day for which bid prices have closed, and expands to include the next trading day at 12:30 AEST [28]. We obtain regional price and quantity forecasts from the MMSDM archive [26]. For each origin we take the latest run whose rows were all published by then and keep the forecast price, total demand, available generation and net interchange for each hour. Runs published between about 04:00 and 12:30 end before the 24th forecast hour; uncovered hours repeat the last covered hour and are flagged (93% of hours are covered). The archive has no run history for October 2022, whose origins are marked uncovered and filled with training-period medians.

Used directly as a forecast, the predispatch price has a test MAE of 147.5,

128 153.3 and 69.4 AUD/MWh in NSW1, QLD1 and TAS1, inflated by spikes that
 129 do not materialize, but a median absolute error of only 17–20 AUD/MWh: it
 130 is informative but must be calibrated.

131 *Price treatments.* We evaluate on raw prices and, as a robustness check that
 132 limits the weight of a few extreme hours in absolute-error metrics, on prices
 133 capped at 650 AUD/MWh, with the cap applied to the price series used as
 134 target and as history, in training and in scoring.

135 4. Method

136 Fig. 1 gives an overview. The forecast is the sum of a linear base on all
 137 inputs, a dual-field residual built from a trend–event decomposition of the
 138 history, and a calibration term, followed by a price floor. Every component is
 139 refit before each test quarter.

140 4.1. Target transformation

141 Prices are mapped by the area hyperbolic sine with median and median-
 142 absolute-deviation scaling [13],

$$y = \operatorname{asinh}\left(\frac{p - \mu}{\sigma}\right), \quad \mu = \operatorname{med}(p^{\text{train}}), \quad \sigma = \frac{\operatorname{med}|p^{\text{train}} - \mu|}{0.6745}, \quad (1)$$

143 and standardized with training statistics. The point forecast is produced in
 144 this space and mapped back by the inverse transform. The quantile forecasts
 145 are produced directly in standardized price space, which keeps their spread
 146 in the units on which they are scored.

147 4.2. Dual-field decomposition of the history

148 Let $x \in \mathbb{R}^{72 \times 2}$ be the standardized history of transformed price and
 149 demand, indexed by $\tau = 1, \dots, 72$.

150 *Event field (DGF).* Events are detected in the price channel $x^{(1)}$ relative
 151 to a robust level. With departures $d_\tau = x_\tau^{(1)} - \operatorname{med}_\tau x_\tau^{(1)}$, we select $K = 8$
 152 positions $c_1, \dots, c_K$ greedily by largest $|d_\tau|$, suppressing positions closer than
 153 three hours to an already selected one. Each event keeps its departure,
 154 soft-thresholded by a learned level $\kappa > 0$, and becomes a Gaussian bump of
 155 learned width w :

$$e(\tau) = \sum_{k=1}^K a_k \exp\left(-\frac{(\tau - c_k)^2}{2w^2}\right), \quad a_k = \operatorname{sign}(d_{c_k}) \max(|d_{c_k}| - \kappa, 0). \quad (2)$$

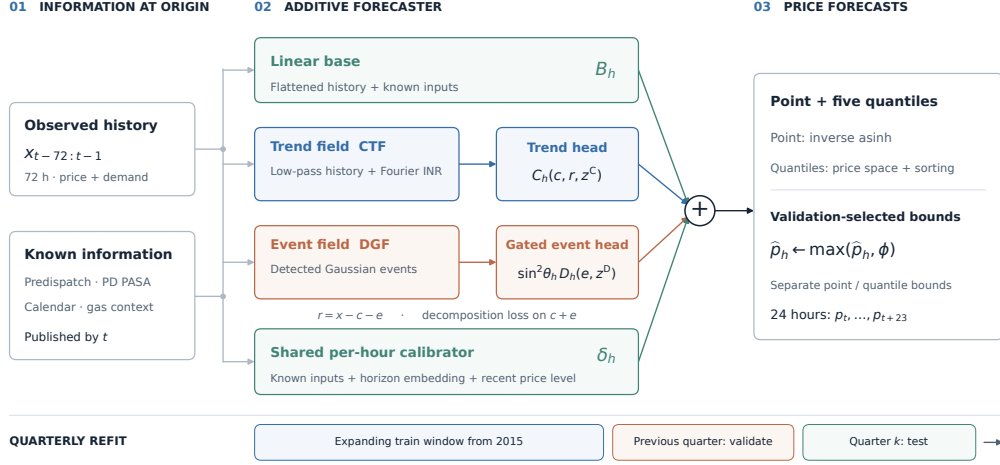


Figure 1: Dual-field forecasting architecture. Historical observations x and information z published by origin t feed the linear base and the appropriate inputs of each branch. The CTF reads low-pass history and time features; the DGF detects price events directly from history. Their heads form a gated residual, with $r = x - c - e$ also read by the CTF head. A shared per-hour calibrator supplies an additional correction. Point outputs are inverse-transformed; price-space quantile outputs are sorted. Separate validation-selected lower bounds are then applied. The bottom strip shows the expanding training window, preceding validation quarter and test quarter. The diagram is schematic; exact feature routing is specified in Section 4.

156 Event positions are determined by the history, while the threshold and
 157 width remain learned parameters ($w = \text{softplus}(w_0) + 0.25$ hours). The
 158 event signal is nonzero only in the price channel; its demand channel is zero.
 159 This constrains the decomposition more directly than globally learned event
 160 positions (Appendix A); it does not impose a fixed event signal throughout
 161 training.

162 *Trend field (CTF).* The CTF is a Fourier-feature implicit neural representa-
 163 tion [8, 6]. With normalized time $\tau' \in [0, 1]$, learned band-limited frequencies
 164 $b_j = \omega_{\max} \text{sigmoid}(\beta_j)$, $j = 1, \dots, 16$, and a low-pass filtered copy $\tilde{x}_\tau$ of the
 165 history,

$$\gamma(\tau') = \left[\cos 2\pi b_j \tau', \sin 2\pi b_j \tau' \right]_{j=1}^{16}, \quad c(\tau) = f_\theta(\gamma(\tau'), \tilde{x}_\tau), \quad (3)$$

166 where f_θ is a small multilayer perceptron. The CTF receives the low-pass
 167 filtered full history, not an event-subtracted input. The decomposition loss
 168 $\mathcal{L}_{\text{dec}} = \|c + e - x\|_2^2 / |x|$ encourages it to account for the signal not represented
 169 by the events, and $r = x - c - e$ is the remainder. The net-load input to its
 170 forecast head is standardized on training data, with logarithmic compression
 171 beyond the training 0.5% and 99.5% quantiles.

172 4.3. Forecast heads, linear base and fusion

173 For each of the 24 forecast hours h , linear heads map field features to a
 174 point forecast in transformed space and to quantiles at the levels 0.05, 0.10,
 175 0.50, 0.90 and 0.95 in price space:

- 176 • *CTF heads* read the trend c , the remainder r , the calendar, the gas
 177 price and the PD PASA net load: the level and its fundamental drivers.
- 178 • *DGF heads* read the event field e , the calendar, the PD PASA spare
 179 capacity and the predispach inputs; the DGF quantile head also reads
 180 the shortfalls of spare capacity below its training 1%, 5% and 10%
 181 quantiles, a piecewise-linear scarcity response.

182 The two are fused by an additive trigonometric gate, $\hat{y}_h^F = \hat{y}_h^C + \sin^2(\theta_h) \hat{y}_h^D$,
 183 with $\theta_h = \frac{\pi}{2} \text{sigmoid}(g_h)$ and g_h linear in the fields, the calendar and the DGF
 184 inputs; each quantile level has its own affine rescaling of g_h , so tail quantiles
 185 can draw on the event field more than the median.

186 The *linear base* $\hat{y}^B$ is one linear map from the standardized, undecomposed
 187 history and all known inputs to the 24 point and quantile outputs. The fused
 188 field forecast is added to it as a residual, so the fields only need to model
 189 what a linear function of the inputs cannot.

190 *Calibrator.* A single perceptron with two 32-unit hidden layers and GELU
 191 activations, shared by all forecast hours, reads hour h 's known inputs (predis-
 192 patch, PD PASA, calendar), a learned four-dimensional horizon embedding
 193 and two transformed-price summaries (last observed value and 24-hour mean),
 194 and outputs additive corrections δ_h to the point and quantile forecasts. Its
 195 output layer is initialized at zero. It lets the model learn a nonlinear response
 196 to predispach, for example that high predispach prices are often not realized.
 197 The point forecast before inverse transformation and the lower bound is

$$\hat{y}_h = \hat{y}_h^B + \hat{y}_h^F + \delta_h. \quad (4)$$

198 *Training.* The recorded training and checkpoint-selection loss is

$$\mathcal{L} = \underbrace{\frac{1}{24} \sum_h (\hat{y}_h - y_h)^2}_{\text{point}} + \underbrace{\frac{1}{24.5} \sum_{h,q} \rho_q (\tilde{p}_h - \hat{\tilde{p}}_{h,q})}_{\text{quantiles}} + \mathcal{L}_{\text{dec}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_g \mathcal{L}_{\text{sparse}}, \quad (5)$$

199 where ρ_q is the pinball loss [29], $\tilde{p}$ is the standardized price, and $\mathcal{L}_{\text{smooth}}$
 200 penalizes squared first differences of the CTF. The retained implementation
 201 computes $\mathcal{L}_{\text{sparse}} = \bar{z} + 10 \max(\bar{z} - 0.3, 0)$, where $z_k = \mathbf{1}[a_k \neq 0]$ and $\bar{z}$ is
 202 the batch mean ($\lambda_s = \lambda_g = 10^{-3}$). This hard indicator has no gradient: the
 203 sparsity term contributes to the reported loss and checkpoint selection, but
 204 does not directly regularize parameter updates. Event sparsity instead results
 205 from selecting at most eight positions and soft-thresholding their amplitudes;
 206 the threshold and width receive gradients through the reconstruction and
 207 forecasting terms. We use AdamW with learning rate 3×10^{-4} , weight decay
 208 10^{-4} , batch size 128 and 30 epochs, and keep the epoch with the lowest
 209 recorded validation loss. The model has 277,747 parameters. Quantiles are
 210 sorted after field fusion, addition of the base, and calibration to prevent
 211 crossings.

212 4.4. Price floor

213 The fitted linear heads can extrapolate to very low prices after the inverse
 214 asinh transform (Section 6.3). We apply an empirical lower bound, $\hat{p}_h \leftarrow$
 215 $\max(\hat{p}_h, \phi)$; this is a forecast calibration choice, not the NEM’s regulatory
 216 market price floor or a physical bound on negative prices. For each refit
 217 and region, ϕ is chosen on that refit’s validation quarter from $\{\text{none}\} \cup$
 218 $\{\min(Q_\alpha(p^{\text{train}}), 0) : \alpha \in \{0.05, 0.1, 0.2, 0.5, 1\}\%$. The training-price sample
 219 here pools the training target windows. Point and quantile lower bounds are
 220 selected separately by mean validation MAE and CRPS over the three seeds,
 221 respectively. No test-quarter prices are used to select the numerical bound.
 222 Its effect on negative-price scores is reported separately from its overall MAE
 223 gain.

224 4.5. Quarterly recalibration

225 Before each of the eight test quarters k (2023Q1 to 2024Q4), every model
 226 is refit on data from January 2015 to three months before the quarter, selects
 227 its checkpoint on those three months, and forecasts the origins of the quarter.
 228 Forecasts of the eight quarters together cover exactly the 17,521 test origins
 229 of each region. Transformations, standardizations and quantile knots are
 230 re-estimated on each refit’s training data.

231 5. Experimental setup

232 *Baselines.* All baselines receive the same history and known inputs, including
 233 predispatch, and are refit on the same quarterly schedule.

- 234 • *XGBoost* [11]: one multi-quantile model per forecast hour on the flat-
 235 tened history, gas price and that hour’s calendar and known inputs
 236 (depth 6, learning rate 0.05, subsampling 0.8, up to 2,000 rounds with
 237 early stopping on validation); the median is the point forecast.
- 238 • *PatchTST* [10] (patch length 16, stride 8, instance normalization, three
 239 layers, $d_{\text{model}} = 128$) and *DLinear* [5] (moving-average decomposition).
 240 Both are channel-independent and would otherwise ignore the known
 241 inputs, so each adds a linear map from the known future inputs and
 242 the demand history to its outputs.
- 243 • *Linear*: one linear map of the price history plus the same linear map of
 244 known inputs, with a mean-squared-error point output and pinball-loss
 245 quantiles; in our experiments it is the strongest linear specification and
 246 dominates DLinear.
- 247 • *Linear base only*: our model without the dual fields, the calibrator and
 248 the floor, trained by the same pipeline; it isolates the contribution of
 249 the structure.

250 Neural baselines output the five quantiles of the transformed price and are
 251 trained with the pinball loss (Adam, batch size 256, 30 epochs, weight decay
 252 10^{-5} , best validation epoch); their learning rates were chosen from $\{10^{-4}, 5 \cdot$
 253 $10^{-4}, 10^{-3}\}$ on validation data. Two further neural baselines, iTransformer
 254 [18] and Informer [17], were less accurate than PatchTST and DLinear in a
 255 static-split screening (Appendix B) and are not refit.

256 *Metrics.* Point forecasts are scored by MAE and by window RMSE, the
 257 root mean squared error of each 24-hour window averaged over windows.
 258 Probabilistic forecasts are scored by the coverage and average interval score
 259 (AIS) of the 5–95% interval and by an approximation of the continuous ranked
 260 probability score [30], twice the mean pinball loss over the five quantiles
 261 (CRPS). MAE is minimized by the median [31]. We also score the tails:
 262 CRPS and the 0.05-quantile pinball loss on hours with negative prices, and
 263 CRPS and the 0.95-quantile pinball loss on hours above 300 AUD/MWh.

264 *Statistical tests.* Each model is trained with seeds 2026, 2027 and 2028. The
265 main table averages separately computed seed scores; tail scores instead
266 evaluate the seed-averaged quantiles. Differences are tested with the Diebold–
267 Mariano test [14] on per-origin losses (MAE averaged over the 24 hours and
268 the seeds; CRPS of seed-averaged quantiles), with a Newey–West long-run
269 variance [32] with 48 lags because hourly origins overlap. Pooled tests use
270 the three-region mean loss at each origin. For tail losses, each origin’s loss is
271 its mean over the event hours in its horizon; pooling averages over regions
272 with defined loss at that origin, and origins without events are dropped. Tail
273 point estimates instead average all event-hour losses within each region before
274 taking a three-region mean. Rolling-origin evaluation and recalibration follow
275 the general recommendations of Tashman [33].

276 *Implementation.* The models are implemented in PyTorch 2.5 and XGBoost
277 3.2 and trained on NVIDIA L40 GPUs. Code and configurations reproduce
278 every table and figure.

279 6. Results

280 6.1. Main comparison

281 Table 2 and Fig. 2 compare the models in the main setting, with quarterly
282 refits and predispatch inputs, and Table 3 tests the differences. The dual-
283 field model reduces the mean MAE of the same linear base from 37.60 to
284 36.64 AUD/MWh on raw prices and from 26.72 to 25.29 on capped prices,
285 and its CRPS from 26.04 to 25.07 and from 14.96 to 13.80 (all $p < 0.01$).
286 It is more accurate than PatchTST, DLinear and the linear model on MAE
287 under both price treatments and in every region ($p < 0.05$), and on capped
288 CRPS ($p < 0.01$); on raw CRPS the differences to these three models are not
289 significant. XGBoost has the lowest MAE and CRPS: our model is behind
290 by 1.11 (raw) and 0.75 (capped) on MAE, most in QLD1 (2.10 raw); in TAS1
291 the raw MAE difference is not significant.

292 6.2. When does the decomposition help?

293 Fig. 3 and Table 4 trace every model across three settings, and the first
294 block of Table 5 isolates the dual fields against the same linear base in each.

295 On the static split (a single fit to 2015–2021), the fields *raise* MAE by 0.92
296 (raw) and 0.95 (capped) ($p < 0.01$), and the dual-field model (41.54) is behind
297 the linear model with the same inputs (40.52). The fields lower the training

Table 2: Main comparison on the 2023–2024 test set (17,521 hourly origins per region, 24-hour horizon). Every model is refit before each test quarter and uses the same inputs, including AEMO predispatch. Cells are means over three seeds; Mean MAE also shows the seed standard deviation. Ours includes the calibrator and the price floor; Linear base only is ours without them and without the dual fields. Lower is better; best in bold.

Model	MAE			Mean MAE	Window RMSE	90% AIS	CRPS
	NSW	QLD	TAS				
<i>Raw prices</i>							
Ours	44.27	38.57	27.07	36.64 ± 0.14	73.69	367.8	25.07
Linear base only	44.86	39.01	28.91	37.60 ± 0.03	74.35	401.2	26.04
XGBoost	43.34	36.47	26.76	35.52 ± 0.02	73.14	347.6	23.11
PatchTST	46.68	40.03	29.84	38.85 ± 0.07	75.94	355.3	24.33
DLinear	46.09	39.54	28.98	38.20 ± 0.09	76.11	338.7	23.62
Linear	45.16	39.51	29.10	37.92 ± 0.02	74.78	339.7	23.66
<i>Capped at 650</i>							
Ours	25.64	25.89	24.35	25.29 ± 0.17	36.07	165.8	13.80
Linear base only	26.77	27.46	25.92	26.72 ± 0.00	38.08	183.8	14.96
XGBoost	25.20	24.78	23.65	24.54 ± 0.02	35.51	160.7	13.10
PatchTST	28.77	28.67	26.87	28.10 ± 0.07	40.00	188.6	15.12
DLinear	29.01	28.44	26.02	27.82 ± 0.05	41.50	185.7	14.96
Linear	27.09	27.84	26.11	27.01 ± 0.01	38.42	186.1	14.98

loss but raise the error on the 2022 validation year: at the selected epoch in NSW1, validation MAE is 60.6 with the fields against 48.6 without. They fit the 2015–2021 level, which shifted in 2022 and again in 2023–2024 (Table 1). Normalizing each history window to remove the level did not help, because volatility shifted as well: the median window standard deviation of QLD1 prices is 2.2 times larger in the test period than in training (Appendix B).

The static-split result is metric-dependent: despite worsening MAE, the fields improve the same base’s CRPS (pooled differences -1.90 raw and -1.88 capped, $p < 0.01$, using seed-averaged quantiles). With quarterly refits, pooled MAE differences are not significant (-0.10 raw, -0.06 capped), while CRPS improves by about 1.1–1.3 ($p < 0.01$). Adding predispatch yields pooled gains in MAE (-0.34 raw, $p < 0.05$; -0.70 capped, $p < 0.01$) and CRPS (-1.13 , -0.94 , $p < 0.01$). These are not uniform regional gains: on raw prices, adding the fields changes MAE by -0.27 in NSW1 ($p = 0.334$), $+0.42$ in QLD1 ($p = 0.009$) and -1.16 in TAS1 ($p < 0.001$). Under capped prices it lowers MAE in all three regions ($p < 0.05$). Predispatch lowers every

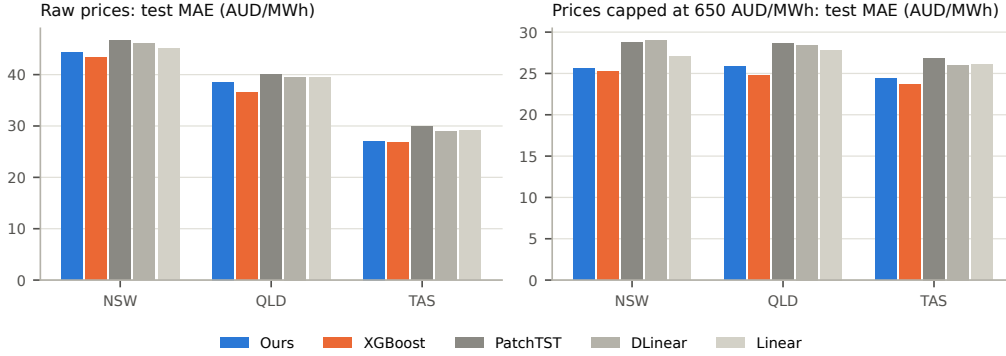


Figure 2: Test MAE per region in the main setting, raw prices (left) and prices capped at 650 AUD/MWh (right); means over three seeds.

314 model’s three-region mean MAE, and XGBoost’s most (39.53 to 35.52 raw),
 315 consistent with useful additional information from the operator’s dispatch
 316 simulation.

317 6.3. Diagnosis and corrections

318 To locate the remaining gap to XGBoost, we split the absolute-error
 319 difference by condition for the calibrated model before applying the lower
 320 bound. In QLD1, its seed-mean MAE gap is 2.9 AUD/MWh; for the seed-
 321 averaged point forecasts used in this diagnosis, the gap is 2.31 (38.56 against
 322 36.25). Hours with a non-positive actual price (13% of hours) account for
 323 47% of the latter gap, and hours in the lowest net-load quintile, largely the
 324 same midday solar hours, for 48%. The calibrated model forecast prices
 325 below -100 AUD/MWh in 0.23% of hours, against 0.01% of actual prices and
 326 none for XGBoost: the 3.2% of hours in which it forecast below -50 carry
 327 31% of the gap. Linear heads extrapolate in asinh space, and the inverse
 328 transform can turn modest transformed-space deviations into deep negative
 329 prices. The fitted XGBoost comparator did not exhibit these extrapolations;
 330 this observation is not a general range-preservation guarantee for boosted
 331 trees.

332 Two corrections follow (second block of Table 5). Letting the model
 333 respond nonlinearly to predispach, by hinge features of the predispach price
 334 or by the calibrator, lowers MAE by 0.25–0.44 ($p < 0.01$ except the raw
 335 calibrator, $p < 0.05$); the calibrator is retained because it also gives the best
 336 tail scores. The validation-chosen price floor lowers MAE by a further 0.31

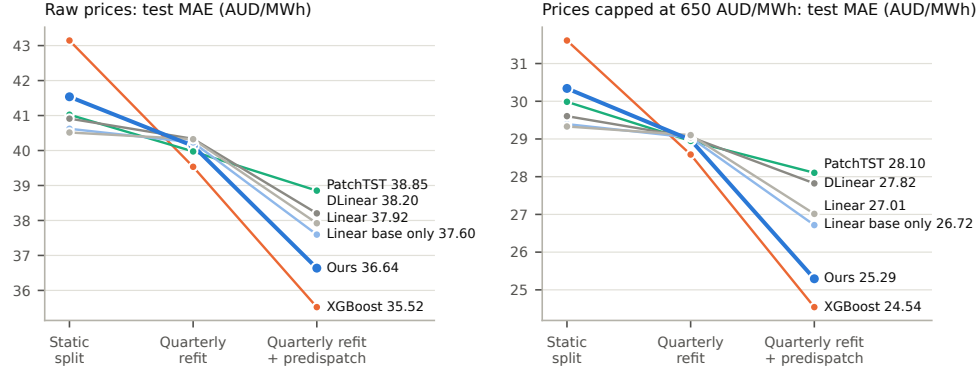
Table 3: Diebold–Mariano tests, ours minus comparator, in the main setting (negative favours ours). Δ columns pool the three regions; NSW, QLD and TAS give the raw-price MAE difference per region. MAE averages losses across seeds; CRPS scores seed-averaged quantiles, so its differences need not equal differences between the seed-mean scores in Table 2. Newey–West variance uses 48 lags. $*p < 0.05$, $**p < 0.01$.

Comparator	Raw prices					Capped at 650	
	Δ MAE	NSW	QLD	TAS	Δ CRPS	Δ MAE	Δ CRPS
Linear base only	-0.96**	-0.59	-0.44	-1.84**	-1.42**	-1.42**	-1.18**
XGBoost	+1.11**	+0.93**	+2.10**	+0.31	+1.11**	+0.75**	+0.31**
PatchTST	-2.22**	-2.41**	-1.46**	-2.77**	-0.20	-2.81**	-1.69**
DLinear	-1.57**	-1.82*	-0.97*	-1.91**	+0.58	-2.53**	-1.46**
Linear	-1.28**	-0.89*	-0.93**	-2.03**	+0.53	-1.72**	-1.49**

337 (raw) and 0.29 (capped), significantly in every region and most in QLD1
338 (-0.82 raw). In QLD1 the chosen point floors are -37 to -48 AUD/MWh
339 in seven of the eight refits. Adding predispatch interconnector flows and
340 limits does not help ($+0.04$ raw, $+0.19$ capped): the predispatch price already
341 reflects the network constraints.

342 6.4. Price tails

343 Table 6 scores the tails. On negative-price hours, our final model has lower
344 CRPS and 0.05-quantile pinball loss than all five comparators in that table
345 under both price treatments, significantly lower in the pooled tests including
346 against XGBoost (raw CRPS 14.52 against 15.96; $p < 0.001$). This does not
347 establish superiority to every development variant: the calibrator without the
348 lower bound achieves raw negative-price CRPS 14.17 and lower-tail pinball
349 loss 4.30, compared with 14.52 and 4.74 after clipping. Capped negative-price
350 CRPS likewise changes from 13.99 to 14.28. The lower bound therefore trades
351 some tail accuracy for lower overall MAE. On capped prices the final model
352 has the lowest spike-hour CRPS in the table (121.1 against 138.3 for XGBoost;
353 $p < 0.01$); its difference to PatchTST is not significant, and PatchTST has
354 slightly lower upper-tail pinball loss. On raw spike hours its CRPS difference
355 to XGBoost is not significant, while it is worse than PatchTST, DLinear and
356 the linear model ($p < 0.05$).



Three-region means over seeds 2026-2028. Ours in the last setting includes the calibrator and the validation-chosen price floor. Axes do not start at zero.

Figure 3: Three-region mean test MAE of every model in three settings: a single fit on 2015–2021, quarterly refits, and quarterly refits with predispatch. Ours in the last setting includes the calibrator and the price floor. Means over three seeds; axes do not start at zero.

357 6.5. Case studies

358 Fig. 4 shows two test days chosen by fixed rules. On the QLD1 day
 359 with the lowest mean price (1 October 2023), the linear base and the dual
 360 fields keep the midday forecast positive; the calibrator brings it into negative
 361 prices but overshoots to about -85 AUD/MWh, and the floor (-43) clips the
 362 overshoot. The day’s MAE is 12.2 for our model, 14.1 for XGBoost and 17.3
 363 for predispatch. On the NSW1 day on which predispatch most overstated
 364 the peak (24 October 2024, predispatch 17,500 AUD/MWh against an actual
 365 price below 300), every model is pulled up at 17:00; ours reaches about 470
 366 and XGBoost about 280 (day MAE 57.1 and 52.8). Such phantom spikes
 367 remain a failure mode.

368 6.6. Computational cost

369 Table 7 reports the size of each model and the wall-clock time of one
 370 quarterly refit (NSW1, 2023Q3 test quarter, seed 2026) on one GPU. Every
 371 model refits in four to six minutes; the dual-field model costs about 20% more
 372 than its linear base and less than PatchTST. A full evaluation (eight refits,
 373 three regions, three seeds) takes a few hours per model on one GPU.

Table 4: Three-region mean test MAE across settings: static split, quarterly refit, quarterly refit with predispatch (PD). Means over three seeds; best per column in bold.

Model	Raw prices			Capped at 650		
	Static	Refit	Refit + PD	Static	Refit	Refit + PD
Ours	41.54	40.13	36.64	30.34	28.98	25.29
Linear base only	40.62	40.23	37.60	29.39	29.04	26.72
XGBoost	43.15	39.53	35.52	31.61	28.59	24.54
PatchTST	41.02	39.97	38.85	29.99	28.95	28.10
DLinear	40.91	40.33	38.20	29.61	29.05	27.82
Linear	40.52	40.32	37.92	29.33	29.10	27.01

374 7. Discussion

375 *Why decomposition has mixed value on a static split.* The identity $c + e +$
376 $r = x$ means the fields introduce no new observations, but their nonlinear
377 representation can express relationships unavailable to a single linear map.
378 In our static-split experiments this added structure worsens MAE while
379 improving the same base’s CRPS. Lower training loss and higher validation
380 MAE are consistent with sensitivity to the changing price regime; they do
381 not isolate regime shift as the sole cause. Informer also performs poorly in
382 the static screening (Appendix B), but differences in normalization alone do
383 not establish the cause of that result.

384 *Value of recalibration and predispatch.* Refitting incorporates more recent ob-
385 servations, and predispatch supplies the operator’s forward-looking assessment
386 of offers, demand and constraints. The pooled ablations support combining
387 these inputs with the dual-field residual, but raw-price MAE gains from the
388 fields are concentrated in TAS1 and do not extend to QLD1. Negative-price
389 scores improve over the main-table baselines for the complete model; those
390 gains cannot be attributed to the decomposition alone, because the calibrator
391 and lower bound also change its forecasts.

392 *The gap to XGBoost.* XGBoost uses these inputs more accurately on average,
393 and its fitted predictions avoid the extreme negative extrapolations observed
394 in the neural model. The calibrator and lower bound recover part of the
395 gap, but XGBoost remains ahead in the 100–300 AUD/MWh band and in

Table 5: Ablations. Pooled Diebold–Mariano differences (negative is better). The first block compares the dual fields with the same linear base in each setting; each row of the second block is compared with the model it modifies. MAE averages seed losses; CRPS scores seed-averaged quantiles. These pooled tests do not imply improvement in every region. $^*p < 0.05$, $^{**}p < 0.01$.

Change	Raw prices		Capped at 650	
	Δ MAE	Δ CRPS	Δ MAE	Δ CRPS
<i>Fields minus linear base</i>				
Static split	+0.92**	-1.90**	+0.95**	-1.88**
Quarterly refit	-0.10	-1.28**	-0.06	-1.06**
Quarterly refit + predispatch	-0.34*	-1.13**	-0.70**	-0.94**
<i>Added to fields + predispatch</i>				
Hinge features on predispatch price	-0.25**	-0.95*	-0.39**	-0.26**
Calibrator	-0.31*	-0.26*	-0.44**	-0.22**
Price floor (after calibrator)	-0.31**	-0.03**	-0.29**	-0.01
Interconnector inputs (after floor)	+0.04	+0.04	+0.19**	-0.04

396 QLD1. For overall MAE on this benchmark, XGBoost is the stronger choice.
397 The dual-field model offers better negative-price scores than the main-table
398 baselines, while the unbounded calibrator variant offers still better negative-
399 price scores. Whether either improves storage or demand-response decisions
400 requires a separate decision-based evaluation.

401 *Limitations.* The study covers three regions of one market and a two-year
402 test period. The empirical lower bound is selected on validation data, but
403 the decision to introduce it followed a diagnosis of test-period errors; these
404 results should therefore be confirmed on a later untouched period. On raw
405 spike hours the model is behind the linear baselines, and its regional gains
406 are heterogeneous. Predispatch run history is missing for October 2022, and
407 its horizon does not always cover all 24 target hours. The hard event-activity
408 penalty has no training gradient. Tail tests drop origins without events, so
409 their 48 lags count retained origins rather than fixed elapsed hours; reported
410 significance is conditional on that weighting and dependence approximation.
411 Hyperparameters were developed on the static split and were not re-tuned
412 for the recalibrated setting.

Table 6: Probabilistic scores in the price tails, main setting, mean of the three regions, scoring seed-averaged quantiles. Lower is better for losses; best among the displayed models is bold. Coverage is evaluated over all test hours. [†]/[‡]: ours is significantly better/worse than that model in a pooled Diebold–Mariano test on per-origin event-hour losses ($p < 0.05$). Point estimates average event-hour losses within each region; tests first average within each origin, so their differences use a different weighting.

Model	Negative-price hours		Spike hours (>300)		
	CRPS	$q_{0.05}$ pinball	CRPS	$q_{0.95}$ pinball	90% coverage
<i>Raw prices</i>					
Ours	14.52	4.74	800.3	675.3	91.5%
Linear base only	16.21 [†]	5.82 [†]	882.3 [†]	777.2 [†]	93.1%
XGBoost	15.96 [†]	6.13 [†]	790.5	640.3	87.7%
PatchTST	17.99 [†]	7.51 [†]	745.2 [‡]	600.3 [‡]	90.0%
DLinear	17.25 [†]	7.33 [†]	740.3[‡]	591.7[‡]	89.1%
Linear	17.25 [†]	7.38 [†]	741.8 [‡]	593.8 [‡]	89.0%
<i>Capped at 650</i>					
Ours	14.28	4.83	121.1	77.2	90.5%
Linear base only	16.23 [†]	6.07 [†]	146.7 [†]	87.5 [†]	92.4%
XGBoost	15.47 [†]	6.13 [†]	138.3 [†]	91.0 [†]	86.2%
PatchTST	17.90 [†]	7.42 [†]	127.8	77.0	90.0%
DLinear	17.19 [†]	7.32 [†]	141.7 [†]	85.5	89.0%
Linear	17.31 [†]	7.44 [†]	141.9 [†]	85.5	88.7%

8. Conclusion

The value of trend–event decomposition in rolling electricity price forecasting depends on the metric, region and recalibration setting. On the static split, adding the fields worsens the same linear base’s MAE but improves its CRPS. With quarterly refits and predispatch inputs, pooled MAE and CRPS improve, although raw-price MAE still worsens in QLD1. The complete model, including the calibrator and empirical lower bound, improves MAE over PatchTST, DLinear and a linear baseline and improves negative-price scores over all five main-table comparators. The lower bound sacrifices some negative-price accuracy relative to the unbounded calibrator. XGBoost retains the lowest overall MAE and CRPS. Future work should test these tradeoffs on an untouched period and in other markets, and model predispatch errors

Table 7: Model size and wall-clock time of one quarterly refit (NSW1, 2023Q3, seed 2026), each run alone on an NVIDIA L40 shared with other users; times are indicative. XGBoost size is the total number of boosting rounds over the 24 per-hour models; Linear base only keeps the field modules, which are still fitted by the decomposition loss.

Model	Parameters	Seconds per refit
Ours	277,747	312
Linear base only	275,629	258
XGBoost	2,871 rounds	269
PatchTST	599,730	364
DLinear	78,120	252
Linear	83,232	237

425 explicitly to handle phantom spikes.

426 **CRediT authorship contribution statement**

427 **zl:** Conceptualization, Methodology, Software, Validation, Formal analysis,
428 Investigation, Data curation, Writing – original draft, Writing – review &
429 editing, Visualization.

430 **Declaration of competing interest**

431 The authors declare that they have no known competing financial interests
432 or personal relationships that could have appeared to influence the work
433 reported in this paper.

434 **Declaration of generative AI and AI-assisted technologies in the** 435 **manuscript preparation process**

436 During the preparation of this work the author(s) used [tool name] in
437 order to [purpose]. After using this tool, the author(s) reviewed and edited
438 the content as needed and take(s) full responsibility for the content of the
439 published article.

Table A.8: Event-to-price mean ratio R_e on historical spike positions in test windows, measured in standardized asinh space.

Event field	NSW	QLD	TAS
Transplanted (Gabor)	-0.03	-0.00	-0.04
Detected events (ours)	0.33	0.33	0.28

440 Data availability

441 All data are public and were obtained from the Australian Energy Market
 442 Operator [25, 26, 27]. Code and configurations that reproduce the results will
 443 be made available.

444 Appendix A. Field separation and static-split development

445 In our forecasting transplant, the original Gabor DGF with globally learned
 446 positions carries little of the historical spike signal, while the CTF absorbs it.
 447 We quantify this by $R_e = \text{mean}_{\mathcal{S}} e^{(1)} / \text{mean}_{\mathcal{S}} x^{(1)}$, where $\mathcal{S}$ contains historical
 448 positions with raw price above 300 AUD/MWh across all test windows. Both
 449 numerator and denominator are in standardized asinh space, and repeated
 450 positions in overlapping windows are retained. The ratio is -0.04 to 0.00 for
 451 the Gabor DGF and 0.28 to 0.33 for detected events (Fig. A.5, Table A.8).
 452 It is a transformed-signal diagnostic, not a fraction of monetary price or
 453 explained variance. Fig. A.6 shows the static-split development, where most
 454 MAE improvement came from target transformation and fundamental inputs.
 455 Table A.9 reports the structural and detector ablations: suppression and the
 456 median level matter, while event count and threshold have smaller effects.

457 Appendix B. Designs that did not help

458 The following were tested with three seeds and are reported for complete-
 459 ness (full records are in the code repository).

- 460 • *Window-normalized fields* on the linear base, with the residual rescaled
 461 by the window’s price scale: raw MAE 44.29 against 41.54 unnormalized
 462 (static split), because test-period volatility is about twice the training
 463 volatility.

Table A.9: Structural and detector ablations of the detected-event DGF on the static split, raw prices, three-region means over three seeds (MAE $\pm$ seed standard deviation). The last column gives the paired MAE difference to the detected-event DGF per seed; positive means the variant is worse.

Variant	MAE	Window RMSE	CRPS	Δ MAE vs detected (seeds)
<i>Structure</i>				
Transplanted dual field (Gabor DGF)	41.88 $\pm$ 0.22	79.21	28.43	+0.38 / +0.03 / +0.58
No decomposition (heads read raw history)	41.90 $\pm$ 0.11	79.31	28.40	+0.30 / +0.25 / +0.51
Adaptive attention atoms + echo	41.84 $\pm$ 0.05	79.01	28.09	+0.30 / +0.28 / +0.30
Detected events + daily echo	42.21 $\pm$ 0.11	79.93	28.22	+0.79 / +0.47 / +0.73
Detected-event DGF	41.54 $\pm$ 0.06	78.80	28.15	–
<i>Detector settings (8 events, 3-hour separation, learned threshold, median)</i>				
4 events	41.65 $\pm$ 0.03	78.85	28.25	+0.13 / +0.01 / +0.18
16 events	41.50 $\pm$ 0.08	78.76	28.15	-0.08 / -0.03 / -0.03
No suppression (1-hour separation)	41.85 $\pm$ 0.03	79.00	28.05	+0.36 / +0.21 / +0.35
6-hour separation	41.51 $\pm$ 0.05	78.77	28.28	-0.05 / -0.11 / +0.05
No threshold	41.63 $\pm$ 0.08	78.98	27.97	+0.21 / -0.03 / +0.07
Mean baseline	41.64 $\pm$ 0.03	78.91	28.02	+0.09 / +0.05 / +0.15

- *Fields on the quantiles only*, with asinh-space quantiles and the point forecast from the linear base: the fields widened the 90% interval from 171 to 418 AUD/MWh and raised CRPS from 28.94 to 31.25 (static split).
- *A future event field*, a three-component mixture for spike and trough probabilities, did not beat a logistic regression on the same inputs in precision–recall.
- *iTransformer and Informer* (static split, raw MAE 43.20 and 46.65) were less accurate than PatchTST (41.02) and DLinear (40.91); Informer’s validation loss was lowest after one to five epochs.
- *Predispatch interconnector inputs*: no gain (Section 6.3).

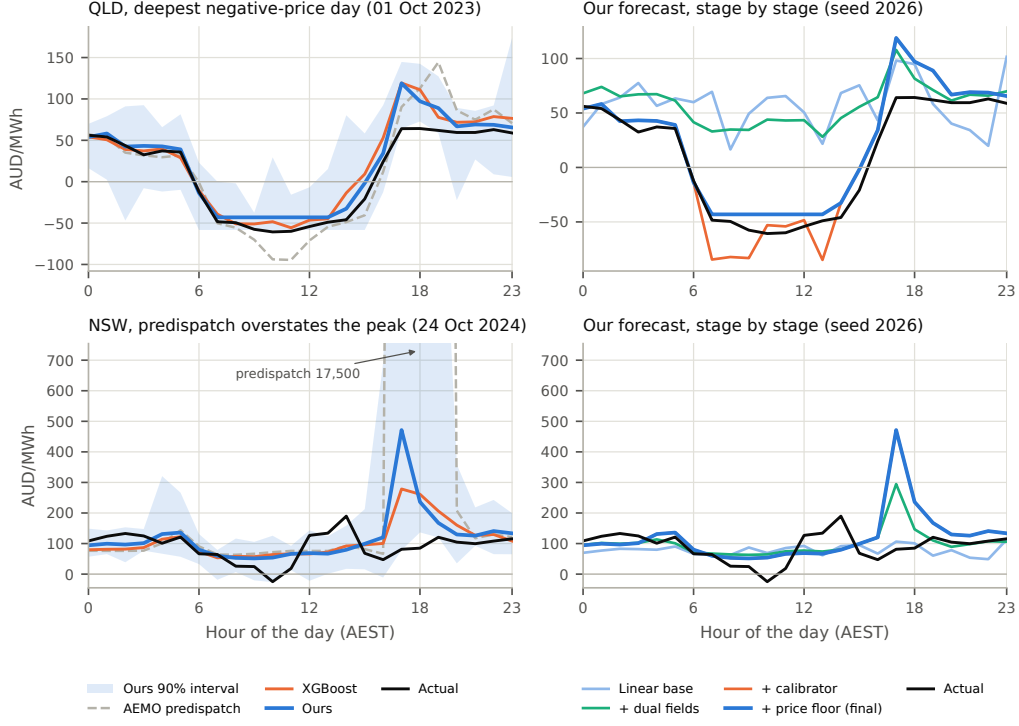


Figure 4: Case studies, chosen by fixed rules on the test split. Top: QLD1 day with the lowest mean price. Bottom: NSW1 day on which predispatch most overstated the peak (forecast above 1,000, actual below 300 AUD/MWh). Left: our forecast with its 90% interval, XGBoost, predispatch and the actual price. Right: cumulative component outputs of the same jointly trained seed-2026 model, not independently retrained ablations. On the NSW1 day the lower bound is not binding, so the calibrated and final forecasts coincide.

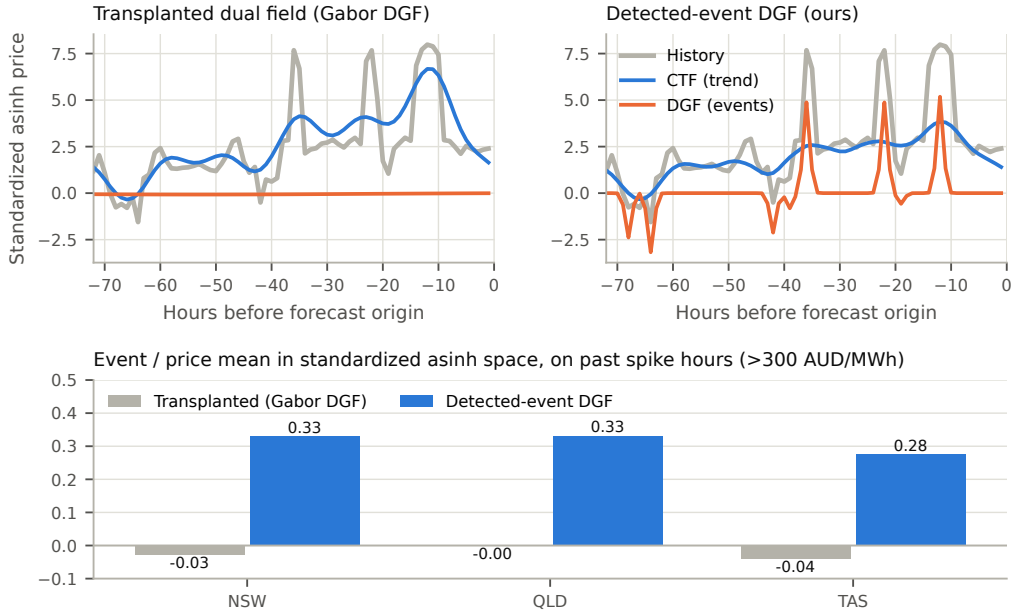


Figure A.5: Field separation. Top: one NSW1 test window in standardized asinh price; the Gabor DGF stays near zero (left), whereas detected events capture localized deviations (right). Bottom: event-to-price mean ratio R_e in standardized asinh space, restricted to historical raw-price spikes above 300 AUD/MWh. This is not a monetary contribution share.

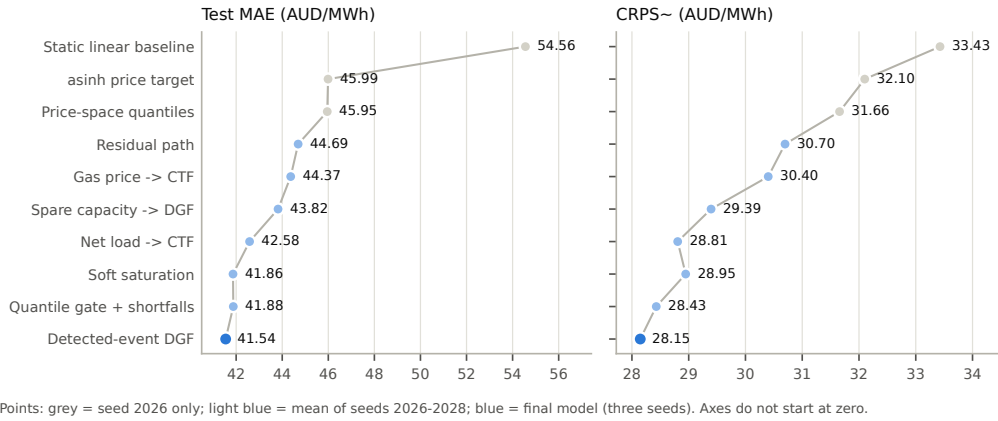


Figure A.6: Development on the static split: three-region mean test MAE and CRPS on raw prices; each step changes one component. Grey: seed 2026 only; blue: means over seeds 2026-2028. Axes do not start at zero.

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